

Joshua Daugherty

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Experience

Software Engineer, Kratos SRE, Inc. | Birmingham, AL

Sep 2025 - Present

- Nondestructive Solutions Development | UltraSpec
- **Note:** Active Secret security clearance. Specific details are confidential under NDA; available to discuss privately.

Lead Python Developer, Celestia Diagnostics | Birmingham, AL

Mar 2024 - Present

- Responsible for designing, building, and maintaining a full-scale acquisition and analysis pipeline (production).
- Architect and deploy cloud infrastructure using AWS Lambda, S3, IAM, and Snowflake for efficient data warehousing and analytics.
- Develop internal GUI applications using a proprietary toolkit to streamline lab workflows and data interaction.
- Build secure, encrypted systems for HIPAA-compliant data storage and transmission across clinical, research, and analytics teams.
- Build, train, and deploy machine learning pipelines optimized for internal diagnostic and research applications.
- Maintain and evolve multiple Python repositories supporting analytics, backend services, and cross-functional data workflows, emphasizing modular and testable design.
- **Note:** Specific details are confidential under NDA; available to discuss privately.

Graduate Teaching Assistant - Probability & Statistics, University of Alabama at Birmingham

Aug 2023 - May 2024

- Teach probability theory and Python programming (Numpy, Matplotlib, and other libraries)
- Assist 200+ graduate and undergraduate students by hosting lectures, office hours, and study sessions
- Proctor exams, prepare lecture materials, grade assignments, and mentor students through collaborative instruction and feedback.

Catalog Specialist, Saucey Inc. | Remote

Aug 2021 - Nov 2022

- Evaluate 100's of stores' onboarding inventory and provide insightful feedback
- Modify SQL and Bash scripts according to parameters on a store-to-store basis
- Quality control of 100's of store menus and website products

Education

3.9/4.0 **Masters in Computer Science**, University of Alabama at Birmingham | Birmingham, AL

3.8/4.0 **Bachelors in Computer Science**, University of Alabama at Birmingham | Birmingham, AL

Relevant Courses: Computer Vision & CNNs, Deep Learning, Artificial Intelligence, Advanced Algorithms & Applications, Software Engineering, Machine Learning, Database Systems, Linear Algebra, Programming Languages, Automata Theory & Formal Languages

Skills

Programming	Python, Java, SQL/PostgreSQL, LaTeX, Bash, REST APIs, PowerShell
Mathematics	Linear Algebra, Differential Equations, Advanced Calculus, Probability Theory, Discrete Mathematics
Software & Tools	AWS (Lambda, S3, EC2, SQS, IAM), Snowflake, Docker, Git, Linux, VS Code, Asana, IntelliJ, MS Office

Projects

Geo Grocer

Full-Stack Mobile-Ready Web App (Personal Project)

- Designed and built a full-stack grocery management app with React (Vite), FastAPI, and SQLite, deployed to production on Railway.
- Implemented geolocation-based store detection, alerting users when they enter a saved store's radius via Web Push notifications and a scheduled reminder system using APScheduler.
- Built a household sharing system allowing users to invite members and collaborate on grocery lists with configurable view-only or edit access.
- Developed touch gesture interactions for mobile (drag-to-reorder, swipe-to-delete, swipe-to-check, long-press inline editing) using dnd-kit and custom pointer event handlers.
- Prepared for native iOS and Android distribution using Capacitor, with guest session support and seamless account conversion.

CUDA-Accelerated Graham Scan

GPU Programming

- Collaborated in a team of three to implement the Graham Scan algorithm for convex hull detection on point clouds.
- Ported algorithm to the GPU using the CUDA Toolkit, optimizing parallelism and memory transfer to reduce runtime on large sets.
- Rendered visual output to display point clouds and resulting convex hulls, comparing CPU and GPU execution paths.
- Performed runtime benchmarking across various data sizes, analyzing speedup and computational efficiency of the GPU version.

CGI Application Tracker – 2nd Place Finish

UA Innovate Hackathon

- Developed a React-based web application with a responsive and interactive front-end to assist CGI's recruitment process.
- Designed a dual-role authentication portal for seamless UI/UX between student applicants and CGI staff.
- Integrated OpenAI API calls to auto-generate resume feedback, enhancing the interface for HR staff insights.

AI Maze-Solving Bot

Automated Reasoning

- Implemented and tuned A* search algorithms for dynamic maze traversal with real-time visual feedback.
- Led effectively in a team environment, collaborating with other students from diverse backgrounds to achieve project goals.
- Identified challenges and devised creative solutions to optimize the bot's performance, making it adaptable to various mazes.

Extracurricular

Association for Computing Machinery (ACM at UAB), *Member*

<https://uabacm.org>

- Attend club events and participate in workshops on a weekly or bi-weekly basis